

CSCE 40103: INTRO TO AI FOR SECURITY

LAB 4: NETWORK ANOMALY DETECTION WITH FLOW RECORDS

Introduction

In this lab, you will apply anomaly detection methods to network flow records. You will train models mostly on benign traffic, score mixed traffic records, and interpret the highest-scoring alerts. The dataset used is ***RT_IOT2022.csv***. Use the label only for evaluation after anomaly scoring.

⊗ **Do not use the label column during unsupervised training.**

Dataset

Item	Requirement
Required file	<i>RT_IOT2022.csv</i>
Dataset source description	 archive.ics.uci.edu  UCI Machine Learning Repository Discover datasets around the world!
Data type	Flow-style network traffic records in tabular form
Target or label column	<code>Attack_type</code> , used only after anomaly scoring for evaluation
Main methods	Gaussian / z-score detection, Isolation Forest, PCA + Isolation Forest
Evaluation	Anomaly scores, alert counts, confusion matrix, classification report, precision, recall, F1-score
Submission file	<code>Lab4-LastName.ipynb</code>

Learning Objectives

- Load and inspect a network flow traffic dataset with benign and attack labels.
- Select meaningful numeric behavior features and remove identifier columns.
- Create a benign training set and a mixed evaluation set for unsupervised anomaly detection.
- Use Gaussian / z-score detection on one flow feature.
- Train Isolation Forest on multiple flow features.
- Use PCA as dimensionality reduction before anomaly detection.
- Evaluate anomaly scores using labels after training.
- Explain why an anomaly score is an investigation lead, not proof of attack.

Required Packages

Use these imports at the beginning of your notebook.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.ensemble import IsolationForest
from sklearn.metrics import confusion_matrix, classification_report
```

Function and Method Parameters Used in This Lab

Function or Method	Important Parameters	Purpose
<code>pd.read_csv()</code>	<code>filepath_or_buffer</code>	Loads the CSV dataset.
<code>df.replace()</code>	<code>[np.inf, -np.inf], np.nan</code>	Replaces infinite values with NaN before cleaning.
<code>df.dropna()</code> / <code>df.fillna()</code>	<code>inplace=True</code> or median value	Removes or fills missing values.
<code>StandardScaler()</code>	No required parameter	Scales numeric features; fit only on benign training data.
<code>PCA()</code>	<code>n_components</code> , <code>random_state=40103</code>	Reduces dimensionality; choose components for $\geq 90\%$ variance.
<code>IsolationForest()</code>	<code>contamination</code> , <code>random_state</code> , <code>n_estimators</code>	Detects anomalies by isolating points in random trees.
<code>confusion_matrix()</code>	<code>y_true</code> , <code>y_pred</code>	Evaluates alert decision against true labels.
<code>classification_report()</code>	<code>y_true</code> , <code>y_pred</code>	Provides precision, recall, and F1-score.

General Instructions

- Create a Notebook named `Lab4-LastName.ipynb` on Google Colab.
- Place ***RT_IOT2022.csv*** in the same folder as your notebook.
- Use markdown headings for each lab step.
- Run every code cell before submitting.
- Include your name, date, course name, and lab number at the top.
- All explanations must be in your own words.
- **No AI use is allowed for this lab.**

Important Cleaning Notes

- Replace `np.inf` and `-np.inf` with `np.nan` before checking for missing values.
- Remove identifier columns such as IP addresses, Flow ID, and timestamp fields, they are not behavior features.
- Keep only numeric columns that describe flow behavior.
- Fill or drop missing values after replacing infinite values; explain your choice in a markdown cell.
- Fit `StandardScaler` only on the benign training set, then transform both the training and evaluation sets separately.
- Do not use the label column at any point during training or scoring, only use it after scoring for evaluation.

Step 1: Create the Notebook

- Create a new Jupyter Notebook named Lab4-LastName.ipynb.
- At the top of the notebook, add a markdown cell with your name, date, course name, and lab number.

Required output: Notebook header cell.

Step 2: Load the Dataset

- Load the dataset into a DataFrame named `df`.
- Display the first five rows.
- Print the shape of the dataset.

Required outputs: First five rows and dataset shape.

Step 3: Inspect Labels

- Display the class counts and class percentages for `Attack_type`.
- Create a binary label for evaluation only:
normal traffic = 0 and attack traffic = 1.

Required outputs: Class counts, class percentages, and class distribution plot.

Step 4: Inspect Columns and Data Types

- Print all column names.
- Display the data type summary.
- Identify which column contains the label.

Required outputs: Column names and data type summary.

Step 5: Clean Missing and Infinite Values

- Replace infinite values with NaN.
- Display the number of missing values in each column.
- Remove rows with missing values or fill missing values using a reasonable method.
- Briefly explain your choice in one markdown cell.

Required outputs: Missing value summary before and after cleaning.

Step 6: Select Numeric Flow Features

- Create a list of numeric feature columns.
- Remove the label column.
- Remove obvious identifier columns if they exist, such as IP address, Flow ID, timestamp, or raw identifier fields.
- Keep only numeric behavior features.

Examples of useful feature types include duration, packets, bytes, flow rate, packet length statistics, inter-arrival time statistics, and TCP flag counts.

Required outputs: Final feature list and number of selected features.

Step 7: Create Benign Training Data and Mixed Evaluation Data

- Create a training set using only benign records.
- Create an evaluation set containing both benign and attack records.
- Do not train unsupervised anomaly detectors on the label.
- Use the label only after scoring to evaluate and interpret results.
- For **RT-IoT2022**, treat normal traffic categories as benign and all attack categories as attack when creating the evaluation labels.

Required outputs: Training set size, evaluation set size, and evaluation label distribution.

Step 8: Scale the Features

- Use `StandardScaler`.
- Fit the scaler only on the benign training features.
- Transform both the benign training set and the mixed evaluation set.

Required outputs: Scaled training and evaluation shapes.

Step 9: Gaussian / Z-Score Detection

- Choose one meaningful numeric feature. Good feature choices include `flow_duration`, `flow_pkts_per_sec`, `payload_bytes_per_second`, `fwd_pkts_tot`, `bwd_pkts_tot`, or `flow_SYN_flag_count`.
- Using benign training data, compute the mean, standard deviation, and z-score for evaluation records.
- Create an alert if the absolute z-score is greater than 3.

Required outputs: Selected feature, mean, standard deviation, top 10 records by absolute z-score, and alert counts.

Step 10: Interpret Gaussian Alerts

- Look at the top 5 Gaussian / z-score alerts.
- For each one, briefly explain which feature made it unusual, whether it was benign or attack according to the label, and one possible reason this method may produce false positives.

Required output: Short interpretation table.

Step 11: Train Isolation Forest

- Train an Isolation Forest model using the scaled benign training data.
- Use the selected numeric features.
- Score the mixed evaluation records.
- Create an anomaly ranking.

Required output: Top 10 records by Isolation Forest anomaly score.

Step 12: Evaluate Isolation Forest

- Use the label only after scoring.
- Create a confusion matrix and classification report using the Isolation Forest alert decision.

Required outputs: Confusion matrix and classification report.

Step 13: Apply PCA for Dimensionality Reduction

- Fit PCA on the scaled benign training data.
- Choose enough components to preserve at least 90% of the variance.
- Transform both training and evaluation records into PCA features.

Required outputs: Explained variance table and number of PCA components selected.

Step 14: Train Isolation Forest on PCA Features

- Train a second Isolation Forest model using the PCA-transformed benign training data.
- Score the PCA-transformed evaluation records.
- Compare the top alerts from Isolation Forest on original scaled features and Isolation Forest on PCA features.

Required outputs: Top 10 PCA-based alerts and short comparison.

Step 15: Final Comparison Table

- Create a final table comparing Gaussian / z-score detection, Isolation Forest, and PCA + Isolation Forest.
- Include number of alerts, precision, recall, F1-score, and short interpretation.

Required output: Final comparison table.

Required Method Comparison Table

Method	Alerts	Precision	Recall	F1-Score	Short Interpretation
Gaussian / Z-Score	?	?	?	?	
Isolation Forest	?	?	?	?	
PCA + Isolation Forest	?	?	?	?	

Step 16: Final Reflection

Answer the following questions in markdown:

- Which method produced the most useful alert ranking?
- Which method was easiest to explain?
- Which method seemed most likely to create false positives?
- Why should anomaly scores be treated as investigation leads rather than proof of attack?

Required output: Short written reflection.

Reference Guides

Anomaly Score Interpretation Guide

Score Type	Plain Meaning	Common Behavior
Z-Score > 3	Feature value is more than 3 standard deviations from the benign mean	Simple, fast; misses multi-feature anomalies.
Isolation Forest score (negative)	More negative score means the point is easier to isolate	Scores all records; lower scores rank as more anomalous.
PCA reconstruction	Anomalies project poorly onto principal components learned from benign data	Useful when features are correlated.
Alert decision	Binary label derived from thresholding the score	Use to compute confusion matrix and F1-score.

Reference Guides

Isolation Forest Parameter Guide

Parameter or Output	Meaning	What to Watch
contamination	Expected fraction of anomalies in the scoring set	Too high may flag many benign records; too low may miss attacks.
n_estimators	Number of isolation trees to build	Higher values give more stable scores; default 100 is usually sufficient.
random_state	Seed for reproducibility	Set to a fixed value so results are consistent across runs.
Anomaly score	Average path length needed to isolate a point	Shorter isolation path = more anomalous.
Decision function output	Signed score: negative = anomaly, positive = normal	Use predict() to get binary labels for evaluation.

What to Submit

- Notebook filename is Lab4-LastName.ipynb.
- Dataset loads correctly.
- All code cells are executed.
- Dataset cleaning and feature selection decisions are explained.
- Gaussian / z-score detection, Isolation Forest, and PCA + Isolation Forest are all included.
- Evaluation uses labels only after scoring.
- Final comparison table and written reflection are included.
- Notebook is clean, organized, and fully executed.

Grading Rubric

Part	Points
Steps 1–4: Notebook setup, data loading, label inspection, column inspection	10
Step 5: Clean missing and infinite values	10
Step 6: Select numeric flow features	10
Step 7: Create benign training and mixed evaluation sets	10
Step 8: Scale the features	5
Steps 9–10: Gaussian / z-score detection and interpretation	15
Steps 11–12: Isolation Forest training and evaluation	15
Steps 13–14: PCA reduction and PCA-based Isolation Forest	10
Step 15: Final comparison table	5
Step 16: Final reflection	5
Comments, explanations, and notebook organization	5
Total	100